

CAD Inspection Tool — User Manual

Table of Contents

1. [Overview](#)
 2. [Getting Started](#)
 3. [Application Layout](#)
 4. [Opening CAD Files](#)
 5. [Navigating the Canvas](#)
 6. [Feature Tree Browser](#)
 7. [Property Inspector](#)
 8. [Registration Groups](#)
 9. [Image Registration](#)
 10. [Camera Capture](#)
 11. [Measurement Queries](#)
 12. [Keyboard Shortcuts Reference](#)
 13. [Supported File Formats](#)
 14. [Troubleshooting](#)
-

1. Overview

CAD Inspection Tool is a metrology-oriented DXF/DWG viewer designed for machine vision alignment and automatic dimension measurement. It allows you to:

- Load and inspect DXF and DWG CAD drawings
- Browse and select individual geometric features
- Capture images from an industrial camera (MindVision)
- Register captured images to CAD geometry for visual alignment

- Define measurement queries for automated inspection

The application uses a dark theme throughout and requires no persistent configuration.

Technology: Built with PySide6 (Qt) and QPainter for high-performance 2D rendering.

2. Getting Started

Launching the Application

```
python main.py
```

To open a file on startup:

```
python main.py path/to/drawing.dxf  
python main.py path/to/drawing.dwg
```

Dependencies

Package	Required	Purpose
PySide6	Yes	GUI framework
ezdxf	Yes	DXF parsing
numpy	Yes	Numerical operations
OpenCV	For registration	Image loading and edge detection
scipy	For ICP registration	Nearest-neighbor search
MindVision SDK	Optional	Industrial camera support

3. Application Layout

```
+-----+
| Menu Bar: [File] [View] [Settings] [Help] |
| Toolbar: [Open DXF] [Import DWG] | [Fit All] | [Pan][Select]|
+-----+
| Feature | | Property |
| Tree | CAD 2D Viewer Canvas | Panel |
| Panel | | |
+-----+
| Status Bar |
+-----+
```

Two additional panels are available as dockable windows:

- **Registration Panel** (View → Registration Panel) — docks on the right
- **Measurement Queries** (View → Query Panel) — docks at the bottom

All panels and the main splitter are resizable by dragging their borders.

4. Opening CAD Files

Opening a DXF File

1. Click **Open DXF** in the toolbar, or choose **File** → **Open DXF...** (Ctrl+O).
2. Select a `.dxf` file from the file dialog.
3. The file is parsed immediately and all features appear in the tree and canvas.

Importing a DWG File

DWG files require a converter backend (ODA File Converter or libredwg) to be installed.

1. Click **Import DWG** in the toolbar, or choose **File** → **Import DWG...** (Ctrl+D).
2. Select a `.dwg` file.
3. A progress dialog shows the conversion stages:
4. Detecting DWG version
5. Running the converter
6. Validating the DXF output
7. Loading features
8. On completion, the converted DXF loads automatically.

If the “Import DWG” button is grayed out, no converter is installed. Install one of:

- **ODA File Converter:** https://www.opendesign.com/guestfiles/oda_file_converter
- **libredwg:** `sudo apt install libredwg-utils`

Then configure the path via **Settings** → **Configure DWG Converter....**

Supported DWG Versions

Version Code	AutoCAD Version
AC1032	2018+
AC1027	2013
AC1024	2010
AC1021	2007
AC1018	2004
AC1015	2000
AC1014	R14

5. Navigating the Canvas

The central area displays the CAD drawing rendered on a dark background with an adaptive grid.

Mouse Controls

Action	Control
Pan	Middle-button drag or right-button drag
Zoom in	Scroll wheel up (1.15x, centered on cursor)
Zoom out	Scroll wheel down (1/1.15x, centered on cursor)
Select feature	Left-click on a feature
Deselect	Left-click on empty area

Canvas Overlays

- **Grid**: Adaptive dotted grid. Spacing adjusts automatically as you zoom.
- **Origin marker**: Red/green crosshair at coordinate (0, 0).
- **Zoom indicator**: Shows current zoom percentage in the bottom-left corner.
- **Feature highlight**: Selected features glow cyan with an outer halo effect.

Fit All

Press **Ctrl+F** or click **Fit All** in the toolbar to fit all geometry into the viewport.

6. Feature Tree Browser

The left panel shows all loaded features organized by type and layer.

Tree Structure

```
Geometry Features
├── Lines (142)
│   ├── LINE [1A2B3C]
│   └── LINE [2C3D4E]
├── Circles (28)
├── Arcs (54)
├── Polylines (12)
└── Splines (8)
Dimensions (36)
Annotations (20)
Layers
├── 0 (210)
└── Outline (50)
Registration Groups
└── Group 1 (5)
```

Searching

Type in the **Filter features...** text box at the top to search features by name. Matching is case-insensitive and filters in real time.

Interacting with Features

- **Click a feature:** Highlights it on the canvas (cyan glow), zooms to fit it, and shows its properties in the Property Panel.
- **Click a group/layer/category node:** Deselects the current feature.
- **Double-click a node:** Expands or collapses it.

Context Menu

Right-click a feature to open the context menu:

- **Add to Group** → **New Group...**: Creates a new registration group and adds the feature.
- **Add to Group** → **[Existing Group]**: Adds the feature to an existing group.

Right-click a group node:

- **Zoom to Group:** Fits the group's bounding box in the canvas.
- **Delete Group:** Removes the group.

Feature Type Colors

Type	Color
Lines	White
Circles	Cyan
Arcs	Yellow
Polylines	Green
Splines	Magenta
Dimensions	Red
Annotations	Gray
Hatches	Dark gray
Points	Orange
Leaders	Amber

7. Property Inspector

The right panel shows details for the currently selected feature.

General Properties

Field	Description
Type	Feature type (LINE, CIRCLE, etc.)
Layer	DXF layer name
DXF Handle	Original DXF entity handle
ID	Internal feature identifier
Color	DXF color index

Geometry Properties (type-dependent)

- **Line:** Start point, end point, length
- **Circle:** Center point, radius
- **Arc:** Center point, radius, start angle, end angle
- **Polyline:** Vertex count, open/closed, extent
- **Spline:** Degree, control points count, fit points count
- **Text:** Text content, position, height, rotation

Measurement Properties

Reserved for future integration with machine vision measurement. Fields: Nominal, Measured, Deviation, Status (PASS/FAIL).

All values are text-selectable for copy-paste.

8. Registration Groups

Registration groups are named collections of CAD features used for image-to-CAD alignment. Each group has a unique color.

Creating Groups

Method 1: Right-click a feature in the tree → **Add to Group** → **New Group...** → enter a name.

Method 2: In the Registration Panel, click **New** → enter a name.

Managing Groups

In the Registration Panel:

- **New:** Create a new group with an auto-generated name.
- **Rename:** Rename the selected group.
- **Delete:** Remove the selected group.

Adding/Removing Features

1. Click a feature in the tree or canvas to highlight it.
2. In the Registration Panel, with the target group selected, click **Add Selected Feature**.
3. To remove a feature, select it in the group's feature list and click **Remove**.

Group Statistics

When a group is selected, the Statistics section shows:

- **Features:** Total count
- **Centroid:** Center point (x, y) of all features
- **Types:** Breakdown by type (e.g., "LINE: 3, CIRCLE: 2")

Group Visualization

Groups are displayed on the canvas as:

- Dashed colored boundary rectangle
 - Semi-transparent colored fill
 - Diamond-shaped centroid marker
 - Group name and feature count labels
-

9. Image Registration

Image registration aligns a captured (or loaded) image with CAD geometry using the features in a registration group.

Prerequisites

1. A CAD file is loaded.
2. A registration group is created with relevant features (circles and lines work best).
3. An image is loaded (from file or camera capture).

Loading an Image from File

1. Open the Registration Panel (**View** → **Registration Panel**).
2. Click **Load Image...**
3. Select an image file (PNG, BMP, or TIFF).
4. Set the **Pixel Size** (mm/pixel) for your imaging system. Default: 0.01 mm/pixel.
5. Click **OK**.

The image appears as a semi-transparent background layer under the CAD geometry.

Registration Workflow

There are three registration modes:

Coarse Registration - Aligns image and CAD by matching centroids and bounding boxes. - Provides a rough initial alignment. Fast, but not pixel-accurate.

Fine Registration (ICP) - Refines the coarse result using Iterative Closest Point. - Requires a successful coarse registration first. - Reports iteration count and convergence status.

Full Registration - Runs coarse then fine in sequence. Recommended for most use cases.

Steps

1. Select a registration group.
2. Load or capture an image.
3. Click **Full Registration**.
4. Check the status label for results (error values in mm, iteration count, convergence).
5. The image shifts to align with the CAD geometry on the canvas.

Interpreting Results

The status label shows:

- **Coarse:** RMSE error in mm (lower is better)
- **Fine:** Iteration count, RMSE error, convergence flag
- **Full:** Coarse error → Fine error, iteration count

A well-registered result typically has a fine error below 0.5 mm.

10. Camera Capture

If a MindVision industrial camera is connected and the SDK is installed, you can capture images directly from the Registration Panel.

Requirements

- MindVision camera (USB or GigE)
- MindVision MVCAM SDK (`LibMVSDK.so`) installed on the system

If the SDK is not installed, the Camera Capture section is hidden and only file-based image loading is available.

Camera Workflow

1. Open the Registration Panel (**View** → **Registration Panel**).
2. The **Camera Capture** section appears at the top of the panel.
3. Click **Refresh** to scan for connected cameras.
4. Select a camera from the dropdown list.
5. Click **Open** — the live preview starts.
6. Adjust camera settings as needed (see below).
7. Click **Capture Frame** — the current frame is frozen and loaded as the image layer.
8. The registration buttons become active. Proceed with registration as described in Section 9.
9. Click **Close** to stop the camera when finished.

Camera Settings

Click the **Settings** button to expand the settings panel:

Parameter	Description
Auto Exposure	Enables automatic exposure control. When on, the exposure slider is disabled.
Exposure	Manual exposure time in microseconds. Range depends on camera hardware.
Gamma	Gamma correction value. Range depends on camera hardware.
Contrast	Contrast adjustment. Range depends on camera hardware.
Analog Gain	Analog gain setting. Higher values increase brightness but may add noise.
Mirror Horizontal	Flips the image left-right.
Mirror Vertical	Flips the image top-bottom.

Settings are applied to the camera hardware immediately when changed.

Re-capturing

You can click **Capture Frame** again at any time while the camera is open. The previous capture is overwritten and the canvas updates with the new image.

11. Measurement Queries

The Query Panel allows you to define and evaluate automated measurement queries against loaded CAD features.

Opening the Query Panel

Choose **View** → **Query Panel**.

Query Language

Enter one query per line. Lines starting with `#` are comments.

Circle center distance:

```
circles(ID1, ID2)
```

Measures the center-to-center distance between two circle features.

Line perpendicular distance:

```
lines(ID1, ID2)
```

Measures the perpendicular distance between two line features.

Feature ID Resolution

Queries can reference features by:

1. Full feature ID (UUID)
2. DXF handle
3. Partial UUID prefix

Example

```
# Measure distance between two mounting holes
circles(1A2B, 3C4D)

# Measure gap between two edges
lines(F5E6, A7B8)
```

Evaluating Queries

Click **Evaluate** to run all queries. Results appear in the table:

Column	Description
Query	The query expression
Value	Measured value in mm
Nominal	CAD nominal value
Deviation	Measured minus nominal
Status	“ok” (green) or error (red)

A summary line shows: “Evaluated: N queries | OK: M | Errors: K”

Saving and Loading

- **Load:** Opens a `.txt` or `.query` file into the editor.
- **Save:** Saves the editor contents to a `.txt` file.
- **Export Results:** Saves evaluation results to `.txt` or `.csv` format.

12. Keyboard Shortcuts Reference

Shortcut	Action
Ctrl+O	Open DXF file
Ctrl+D	Import DWG file
Ctrl+F	Fit All (zoom to fit)
Ctrl+Q	Exit application
Scroll wheel	Zoom in/out (centered on cursor)
Middle/right drag	Pan
Left click	Select feature
Right-click (tree)	Context menu

13. Supported File Formats

Input Formats

Format	Extension	Notes
DXF	<code>.dxf</code>	Native support via ezdxf
DWG	<code>.dwg</code>	Converted to DXF via ODA or libredwg
Images	<code>.png</code> , <code>.bmp</code> , <code>.tif</code> , <code>.tiff</code>	Loaded as registration image layer

Supported DXF Entity Types

Entity	Rendered	Notes
LINE	Yes	
CIRCLE	Yes	
ARC	Yes	Circular and elliptical arcs
ELLIPSE	Yes	Rendered as elliptical arc
POLYLINE / LWPOLYLINE	Yes	Open, closed, and filled
SPLINE	Yes	Evaluated with ezdxf BSpline engine
TEXT / MTEXT	Yes	With formatting code conversion
DIMENSION	Yes	Decomposed into lines, solids, text
HATCH	Yes	All edge types supported
POINT	Yes	
LEADER	Yes	With arrowhead
INSERT	Decomposed	Block references exploded into child entities

Output Formats

Format	Extension	Notes
Query results (text)	.txt	Human-readable results
Query results (CSV)	.csv	Structured data export

14. Troubleshooting

“Import DWG” button is disabled

No DWG converter backend is detected. Install ODA File Converter or libredwg, then configure via **Settings** → **Configure DWG Converter...**

DWG conversion fails

- Verify the DWG file is not corrupted (try opening in a CAD viewer).
- Check that the converter executable is in your PATH or configured correctly.
- For large files, the 300-second timeout may be insufficient. Contact support.

Canvas appears blank after loading a file

- Click **Fit All** (Ctrl+F) to reset the viewport.
- The file may contain features at very different coordinate ranges. Check the Feature Tree to verify features were loaded.

Camera not detected

- Ensure the MindVision SDK (`libMVSDK.so`) is installed.
- Verify the camera is connected (USB cable or Ethernet for GigE cameras).
- Click **Refresh** in the Camera Capture section.
- For GigE cameras, ensure the camera and computer are on the same network subnet.

Registration produces poor alignment

- Ensure the registration group contains relevant features (circles and lines are most effective).
- Verify the **Pixel Size** is correct for your imaging system.
- Check that the captured image shows the correct area of the part.
- Try adding more features to the registration group.
- The image should contain clear edges corresponding to the CAD features in the group.

Application crashes on DWG import

This was a known issue where DWG conversion callbacks executed GUI operations from a background thread. Ensure you are running the latest version where this has been fixed.